

Migration Shocks and Rent Growth in Supply-Constrained Housing Markets

STAT 274 Final Project Empirical Draft

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Abstract

This project studies whether pandemic-era migration shocks were associated with faster local rent growth, and whether the relationship was stronger in housing markets with tighter supply. I build a county-year panel for 707 U.S. counties over 2019–2023, combining Zillow’s county-level Zillow Observed Rent Index (ZORI), Census county population estimates, Census Building Permits Survey data, and county ACS controls from Census Reporter. The treatment is net migration as a share of lagged population, and the outcome is annual log rent growth. In a two-way fixed effects model, a one percentage point increase in the net migration rate is associated with a 0.71 percentage point increase in annual rent growth; adding time-varying controls gives 0.75 percentage points. A random-forest double machine learning estimate is smaller, 0.41 percentage points, with a confidence interval that includes zero. The evidence is consistent with migration demand pressure mattering for rents, while heterogeneity by supply and permitting constraints is weaker and should be interpreted cautiously.

1 Motivation and Causal Question

The COVID-19 pandemic weakened the link between workplace location and residential location for many households. At the same time, rents rose sharply in many Sun Belt, high-amenity, and lower-density markets. This project asks whether places receiving larger migration inflows experienced faster rent growth, and whether this relationship was larger in markets where housing supply was more constrained.

The main causal question is:

What is the effect of pandemic-era net migration inflows on annual local rent growth?

The heterogeneity question is:

Are rent effects larger in supply-constrained counties?

I treat net migration as the treatment of interest. Remote work is an important background mechanism, but this draft does not directly estimate the effect of remote-work adoption. Instead, it asks whether realized county-level population inflows translated into rent pressure. This framing is more empirically tractable because migration is measured at the county-year level and can be linked directly to local rent outcomes.

2 Data and Variable Construction

The empirical panel uses publicly available data sources. The outcome comes from Zillow’s public research data page, which provides the ZORI county time series as a downloadable CSV and

describes ZORI as a repeat-rent index for listed rental units.¹ I aggregate monthly ZORI to county-year averages, requiring at least six non-missing monthly observations in a year. The outcome is annual log rent growth,

$$Y_{ct} = \Delta \log(\text{ZORI}_{ct}).$$

The treatment is the Census Population Estimates Program’s annual net migration count divided by lagged county population.² This is closer to the proposal’s migration-shock concept than using total population growth because it subtracts natural increase from births and deaths:

$$D_{ct} = \frac{\text{Net migration}_{ct}}{\text{Population}_{c,t-1}}.$$

County characteristics come from Census Reporter’s public ACS interface.³ These include median household income, median gross rent, housing units, vacancy rates, renter share, and unemployment. To better capture the zoning and regulatory channel in the proposal, I also add annual county building permits from the Census Building Permits Survey.⁴ Direct zoning ordinances are not available in a complete county-year panel, so I use pre-pandemic permits per 1,000 residents as an observable proxy for permitting restrictiveness and supply elasticity. The supply-constraint index averages standardized high median gross rent, high renter share, low vacancy rate, and low pre-pandemic permitting.

Appendix Table A1 summarizes the final sample. Zillow county coverage is not universal; after merging sources and dropping missing values, the sample contains 2,912 county-year observations from 707 counties. Appendix Figures A1 and A2 show the two main descriptive patterns: average rent growth rose sharply in 2021–2022, and county-years with higher net migration rates tend to have higher average rent growth.

3 Identification and Estimation

The main challenge is endogenous migration. Counties with improving labor markets, changing amenities, or local policy shocks may attract migrants and experience rent growth for reasons unrelated to migration itself. The baseline specification uses county and year fixed effects:

$$\Delta \log(\text{Rent}_{ct}) = \beta D_{ct} + X'_{ct}\gamma + \alpha_c + \delta_t + \varepsilon_{ct}.$$

County fixed effects remove time-invariant local differences, while year fixed effects absorb national shocks such as inflation, mortgage-rate changes, and broad pandemic housing demand. The controlled fixed effects model includes county population, annual population growth, and lagged ZORI rent. The identifying assumption is a conditional parallel-trends-style claim: after fixed effects and controls, counties with larger net migration shocks would otherwise have had similar rent growth to counties with smaller shocks.

¹<https://www.zillow.com/research/data/>; county ZORI CSV used by the script: https://files.zillowstatic.com/research/public_csvs/zori/County_zori_uc_sfrcondomfr_sm_sa_month.csv.

²Census county population estimates: <https://www2.census.gov/programs-surveys/popest/datasets/2010-2019/counties/totals/co-est2019-alldata.csv> and <https://www2.census.gov/programs-surveys/popest/datasets/2020-2023/counties/totals/co-est2023-alldata.csv>.

³<https://api.censusreporter.org/1.0/data/show/latest>.

⁴Census Building Permits Survey county files: <https://www2.census.gov/econ/bps/County/>. The analysis uses the December year-to-date county files for 2018–2023.

For heterogeneity, I estimate

$$\Delta \log(\text{Rent}_{ct}) = \beta D_{ct} + \theta D_{ct} S_c + X'_{ct} \gamma + \alpha_c + \delta_t + \varepsilon_{ct},$$

where S_c is the standardized supply-constraint index. A positive θ means migration inflows are associated with larger rent increases in more constrained counties.

To relax functional-form assumptions, I also estimate a partially linear double machine learning model:

$$Y_{ct} = \theta D_{ct} + g(W_{ct}) + u_{ct}, \quad D_{ct} = m(W_{ct}) + v_{ct}.$$

The nuisance covariates W_{ct} include year indicators, population, population growth, lagged rent, ACS income, ACS median gross rent, unemployment, vacancy, renter share, pre-pandemic permits, and the supply-constraint index. The main nuisance learner is a random forest with five-fold cross-fitting. I also compute lasso and gradient boosting estimates as sensitivity checks. Inference uses county-clustered standard errors.

4 Results

Appendix Table A2 reports the main estimates. Coefficients are scaled as percentage-point changes in annual rent growth from a one percentage point increase in the net migration rate. The fixed effects estimates are positive and statistically distinguishable from zero. The simplest two-way fixed effects estimate is 0.71, and the controlled fixed effects estimate is 0.75. Interpreted literally, a county with a one percentage point higher net migration rate has roughly 0.7–0.8 percentage points faster annual rent growth.

The random-forest DML estimate is smaller: 0.41 percentage points, with a 95 percent confidence interval from -0.10 to 0.93. The trimmed-overlap DML estimate is larger at 1.22 percentage points. This difference suggests that the average effect is sensitive to how much weight the estimator places on counties with extreme residualized migration shocks. The interquartile range of the net migration rate is 1.11 percentage points, so the main DML estimate implies that moving from the 25th to the 75th percentile of migration pressure is associated with about 0.46 percentage points higher annual rent growth.

Appendix Figure A3 visualizes the main estimates and two additional DML sensitivity checks. The fixed effects estimates are tightly positive. The random-forest DML and lasso DML estimates remain positive but smaller, while the boosting DML estimate is close to zero and imprecise. The sign is generally stable, but the machine-learning adjusted magnitude is model-dependent.

Appendix Table A3 examines heterogeneity by supply and permitting constraints. The interaction is positive but imprecise after adding low pre-pandemic permitting to the constraint index. Appendix Figure A4 shows the same idea descriptively. In the top migration quartile, more constrained counties have visibly higher average rent growth than less constrained counties, but the regression interaction confidence interval includes zero. I therefore treat the zoning/permitting heterogeneity evidence as weakly suggestive rather than decisive.

5 Diagnostics, Limitations, and Conclusion

The DML nuisance models predict the outcome with a cross-fitted RMSE of 0.0236 log points. The treatment model has an out-of-fold R^2 of 0.936, meaning that observed county characteristics and year indicators explain a large share of net migration variation. This is useful for adjustment,

but it also warns that overlap may be limited: counties with very high inflows are systematically different from low-inflow counties.

The most important limitation is identification. Net migration is measured directly, but it is not randomly assigned. Local economic shocks, changing amenities, or expected rent appreciation could affect both migration and rent growth. Zillow also provides strong high-frequency market rent data only for counties with enough rental listings, so the sample is more urban and rental-market oriented than the full universe of U.S. counties. ACS controls are currently used as a cross-sectional county snapshot rather than as a full annual panel. Finally, permits are only a proxy for zoning: low permitting may reflect regulation, limited land, weak demand, or administrative reporting differences.

Overall, the evidence supports the core hypothesis of the proposal: migration-driven demand pressure appears to matter for local rents. The fixed effects estimates imply that a one percentage point increase in the net migration rate corresponds to about 0.7–0.8 percentage points faster annual rent growth. A random-forest DML estimate is smaller and less precise, but still positive. The zoning channel is represented with a permitting-based proxy rather than direct ordinance data, and the estimated heterogeneity by this expanded constraint index is positive but not precise. The strongest next improvement would be to add a direct land-use regulation measure, such as a Wharton-style regulatory index, and sharper quasi-experimental variation from pre-pandemic remote-work exposure interacted with pandemic timing.

A Tables and Figures

Variable	Mean	SD	P25	Median	P75
Annual rent growth (pct. points)	6.50	3.86	3.78	5.65	8.99
Net migration rate (pct. points)	0.51	1.29	-0.16	0.26	0.95
Mean annual ZORI rent (dollars)	1443.01	520.95	1111.99	1351.73	1658.00
Census population estimate	438733.34	675321.08	142142.50	232477.50	483185.25
Median household income	84629.47	20978.19	69473.00	80266.00	96227.00
Vacancy rate	0.09	0.05	0.05	0.08	0.11
Renter share	0.33	0.10	0.25	0.32	0.38
Pre-pandemic permits per 1,000 residents	3.20	3.98	0.00	1.97	5.20

Table A1: Descriptive statistics for the estimation panel.

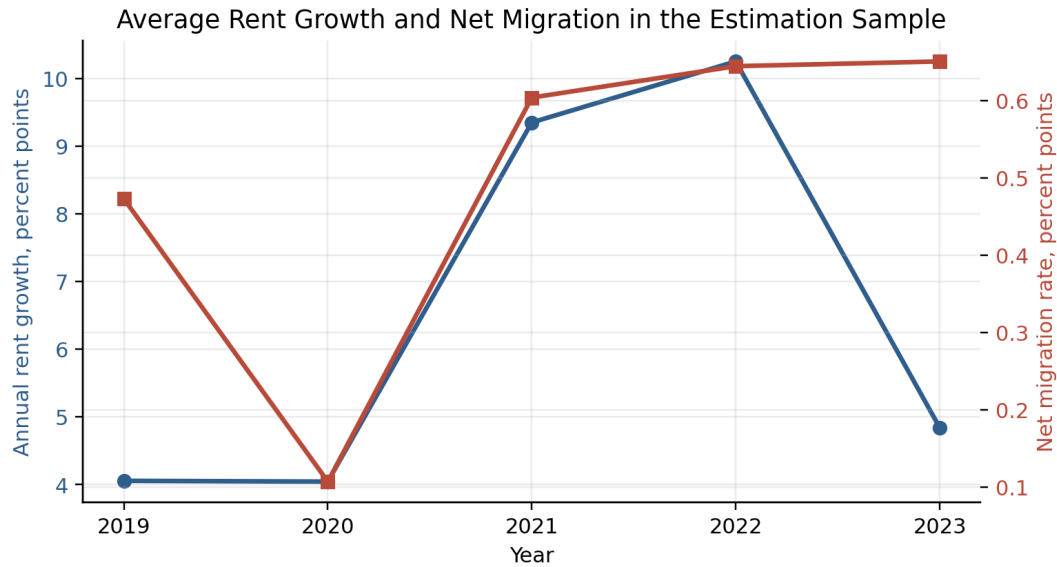


Figure A1: Average annual rent growth and net migration rate in the estimation sample.

	FE	FE + Controls	DML RF	DML RF Trimmed
Net migration rate	0.71	0.75	0.41	1.22
95% CI	[0.31, 1.11]	[0.35, 1.16]	[-0.10, 0.93]	[0.79, 1.65]
Observations	2,912	2,912	2,912	2,794
Counties	707	707	707	699
County FE	Yes	Yes	No	No
Year FE	Yes	Yes	Yes	Yes
Controls	No	Time-varying	Flexible	Flexible

Table A2: Estimated effect of net migration on annual log rent growth.

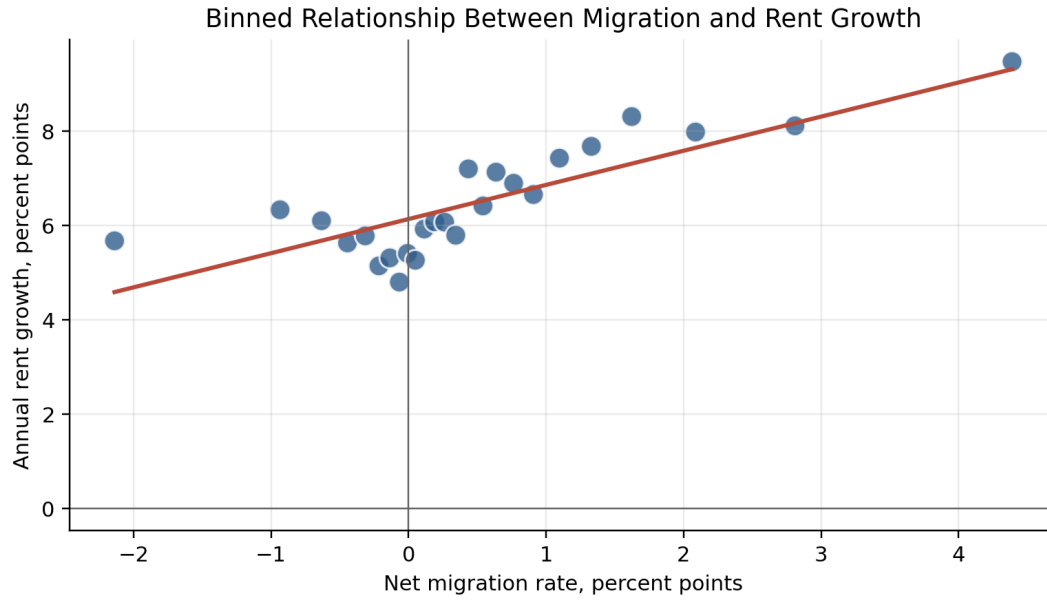


Figure A2: Binned relationship between net migration and annual rent growth.

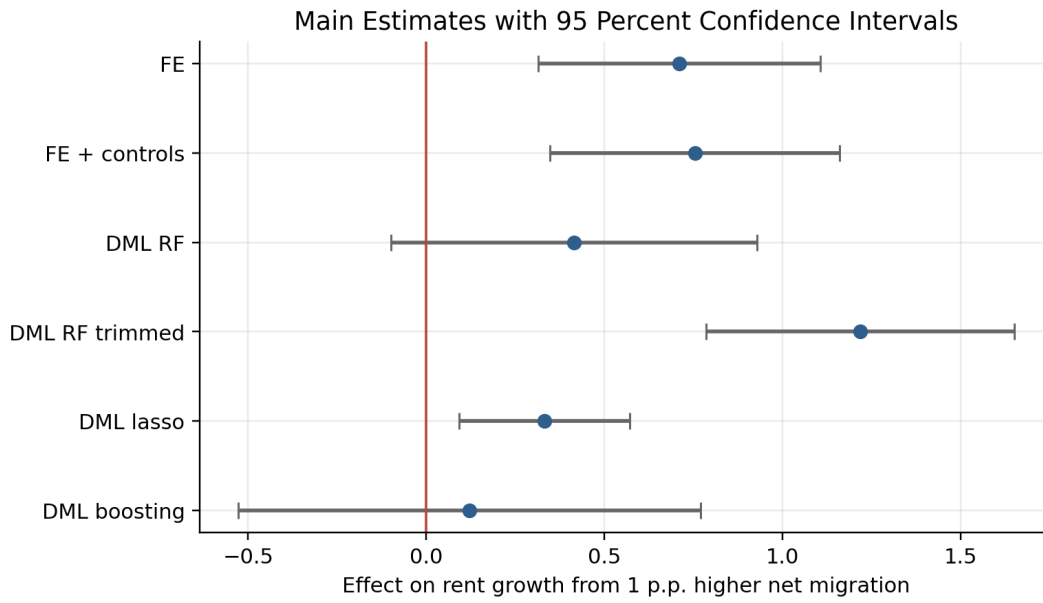


Figure A3: Main effect estimates with 95 percent confidence intervals.

	Estimate	95% CI
Net migration rate	0.76	[0.36, 1.16]
Net migration rate $\times$ supply constraint	0.10	[-0.37, 0.56]
Observations	2,912	
County FE	Yes	
Year FE	Yes	

Table A3: Heterogeneity by the supply-constraint index.

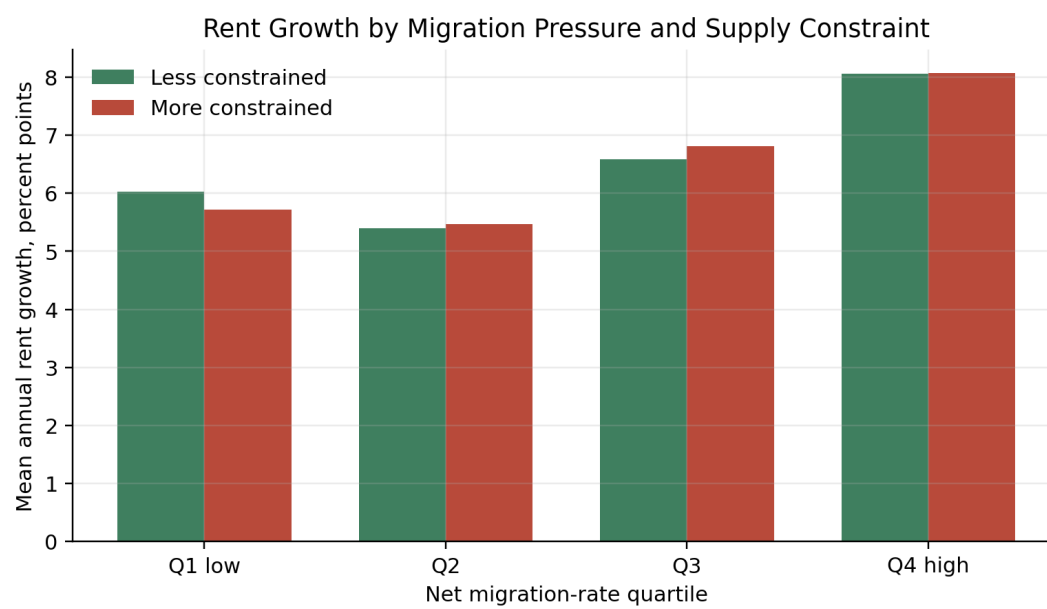


Figure A4: Mean rent growth by migration-pressure quartile and supply-constraint group.

B Reproducibility Notes

The file `analysis.py` downloads all public inputs, constructs the analysis panel, estimates the fixed effects and DML models, generates the figures, and writes the generated LaTeX numbers used by the report. The tables can be refreshed by rerunning:

```
.venv/bin/python analysis.py
```

C Data Sources

Dataset	Variables Used	Role
Zillow ZORI county CSV	Monthly county rent index	Primary outcome after annual aggregation
Census county population estimates	Population and net migration	Treatment and population controls
Census Reporter ACS API	Income, gross rent, housing units, vacancy, renter share, unemployment	DML controls and supply-constraint index
Census Building Permits Survey	Annual housing units authorized by permits	Permitting and zoning proxy in supply-constraint index

Table A4: Public data sources used in this empirical draft.